Steven Thao

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Education

McGill University | **GPA: 3.97/4.0** — Montreal, Canada
B.Eng. in Computer Engineering — Expected Dec. 2027
Dean's Honours List, 2024–2025
In-course scholarships (C$3,500 total): J B Woodyatt; Kenneth and Helen Porter

Experience

Stingray | *Full-Stack Developer Intern (Part-time)* — Sep. 2026 - Present
- Cut **PostgreSQL** query time for retrieving conversations used in metrics from **30 ms to 15 ms** through indexing.
- Refined **AI-generated Google review reply drafts** for business customers by comparing models, temperature settings, and prompt context on a fixed set of **100 reviews**, manually evaluating adherence to predefined personas and response criteria.
- Implemented coachmark onboarding tutorials and deferred feature data fetching until users opened a dropdown.

Full-Stack Developer Intern — Sep. 2025 - Dec. 2025
- Accelerated conversation retrieval queries by up to **7x** through **SQL** indexing and batching in **Django**.
- Reduced retrieval time by **80%** for a customer-facing RAG chatbot through embedding optimizations and **HNSW** indexing with **pgvector**.
- Built frontend features in **React** and **TypeScript**, updating legacy components, state management, and user-facing workflows.

OneSpan | *Backend Engineer Intern* — May 2026 - Aug. 2026
- Developed a **Java/Spring Boot migration tool** used by the backend team to migrate **1000s of production customer records**, replacing manual record creation.
- Built a **GitLab CI pipeline** to deploy and run the migration tool on demand in **Docker** containers on **AWS EKS**.
- Resolved **40+ tickets** in OneSpan Sign's backend, including dependency upgrades across **five Java/Spring Boot microservices** and JUnit migrations requiring test-code and configuration changes.

McGill Rocket Team | *Radio Systems Team Member* — Sep. 2025 - May 2026
- Implemented **C++ MQTT** publishing on **Teensy 4.0** microcontrollers for a one-way rocket telemetry link, including connection-status handling.

McGill AI Lab | *Machine Learning Project Lead* — Nov. 2024 - May 2025
- Led **12 members** through task assignment and code review; implemented binary classifiers and neural networks in **NumPy**.
- Evaluated DNNs on MNIST and Fashion-MNIST, achieving **up to 98% test accuracy**.
- Authored a **50+ page** technical reference covering mathematical foundations, implementation strategies, and optimization techniques for lab members.

Projects

Sign Sync – ASL-to-Speech Prototype | Devpost — Nov. 2025
- Integrated a **CNN** visual recognition model for a fixed vocabulary of ASL signs into a Python application for speech output in virtual meetings.
- Contributed **GPT**-powered language augmentation code to convert recognized symbols into words and expressions.

AutoBot – AI Car Sales Chatbot | Devpost — Nov. 2024
Winner, Matador AI LLM Challenge – McGill CodeJam 14
- Implemented the **RAG** retrieval pipeline for a dealership chatbot using embeddings, vector database storage, and **cosine similarity** search.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C, C++, Bash, HTML/CSS, VHDL
Frameworks & Libraries: Spring Boot, Django, React, JUnit, NumPy, TensorFlow
Databases & Cloud: PostgreSQL, Redis, pgvector, AWS (EKS)
Developer Tools: Git, GitLab CI, Docker, Argo CD